Microeconomics I

Class 8 — Choice under uncertainty: expected utility and risk aversion

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Plan for today

Warm-up; two choices, by hands up

Lotteries and the axioms

Expected utility: form and theorem

In pairs: reduce and rank

Break

Your two choices, revisited; uniqueness

Money lotteries and risk aversion

Certainty equivalent and risk premium

In pairs: one game, three people

Summary

Practicalities

- Problem set 2: handed in at the start of class, Wed 21 Oct; returned graded on Tue 27 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: always true, or not?

True for every consumer?

p_1 rises:
own substitution effect ≤ 0

Inferior good
$\Rightarrow$ Giffen good

compensating variation (CV)
$=$ equivalent variation (EV)

$$\frac{\partial h_1}{\partial p_2} = \frac{\partial h_2}{\partial p_1}$$

Class 7, slides 9, 12, 16, 20. Mas-Colell, Whinston and Green (MWG) Prop. 3.G.2, 3.G.3; (3.I.1)–(3.I.2).

Two choices: which would you take?

Imagine the prizes are real and paid once. No right answer.

First choice

A: 1 million euros for sure

B: 10 %: 5 million euros
89 %: 1 million euros
1 %: nothing

Second choice

C: 11 %: 1 million euros
89 %: nothing

D: 10 %: 5 million euros
90 %: nothing

Hands up: A or B? Then: C or D?

A classic pair of choices; the source in class.

Lotteries and expected utility

MWG 6.B

A simple lottery is a list of probabilities: a point in a triangle

Simple lottery: $L = (p_1, \ldots, p_N)$, $p_n \geq 0$, $\sum_n p_n = 1$; p_n = probability of outcome n.

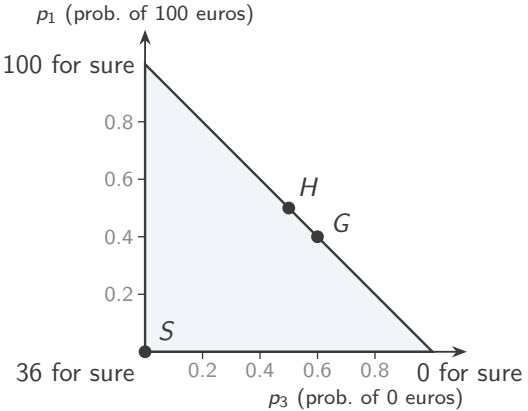
Outcomes: 100, 36, 0 euros.

$S = (0, 1, 0)$: 36 euros for sure

$H = (0.5, 0, 0.5)$: a coin, 100 or 0

$G = (0.4, 0, 0.6)$: 40 % chance of 100

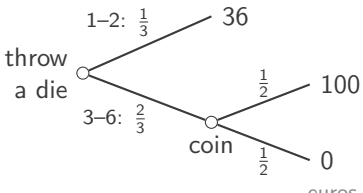
MWG Def. 6.B.1, p. 168; Fig. 6.B.1, p. 169 draws the same set as an equilateral triangle.



$p_2 = 1 - p_1 - p_3$: 0 on the hypotenuse, 1 at S

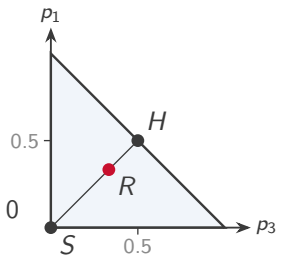
A compound lottery reduces to a simple one: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Compound lottery $(L_1, \ldots, L_K; \alpha_1, \ldots, \alpha_K)$: you get L_k with probability α_k.
Reduced lottery: $\alpha_1 L_1 + \cdots + \alpha_K L_K$.



$R = (S, H;\ \frac{1}{3}, \frac{2}{3})$

$\frac{1}{3}(0,1,0) + \frac{2}{3}(0.5,0,0.5)$
$= (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
= one throw: 1–2 → 100,
3–4 → 36, 5–6 → 0



Only the reduced lottery matters (MWG's consequentialist premise): the die-then-coin tree and one throw are the same lottery.

MWG Def. 6.B.2, pp. 169–170; Figs. 6.B.2–6.B.3, p. 170.

Rational and continuous preferences over lotteries

Rational: complete, transitive (class 1).

Continuous: for all $L, L', L'' \in \mathcal{L}$ the sets $\{\alpha \in [0,1] : \alpha L + (1-\alpha)L' \succsim L''\}$ and $\{\alpha \in [0,1] : L'' \succsim \alpha L + (1-\alpha)L'\}$ are closed.

Small probability changes never reverse a strict ranking.

> **Fails: safety first** (no chance of 0 beats any chance).
> $L = 100$, $L' = 0$, $L'' = 36$ euros for sure: second set $= [0, 1)$.

MWG Def. 6.B.3, p. 171 ("safety first": same page). Class 3: lexicographic order, not continuous.

Independence: mixing both with the same L'' keeps their ranking

Independence axiom: for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0,1)$:

$$L \succsim L' \iff \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$



$\alpha L + (1-\alpha)L''$ $\qquad$ $\alpha L' + (1-\alpha)L''$

Tails gives both the same L'', and you get it instead of L or L', never with it: unlike bundles of goods.

MWG Def. 6.B.4, p. 171 (fn. 5: von Neumann and Morgenstern, 1944); coin-toss reading and Fig. 6.B.4, p. 172.

Expected utility: $U(L) = u_1 p_1 + \cdots + u_N p_N$ is linear in probabilities

U has the expected utility form if there are numbers $u_1, \ldots, u_N$ with $U(L) = u_1 p_1 + \cdots + u_N p_N$ for every L.

Such a U is a von Neumann–Morgenstern (vNM) expected utility function.

Her utilities for 100, 36, 0 euros: 10, 6, 0.

$U(S) = 6, \quad U(H) = 5$

$U(R) = \frac{1}{3} \cdot 10 + \frac{1}{3} \cdot 6 + \frac{1}{3} \cdot 0 = \frac{16}{3} = \frac{1}{3}\, U(S) + \frac{2}{3}\, U(H)$

MWG Def. 6.B.5 and Prop. 6.B.1 (expected utility form $\iff$ U linear), p. 173.

Continuity + independence ⇒ expected utility: straight, parallel lines

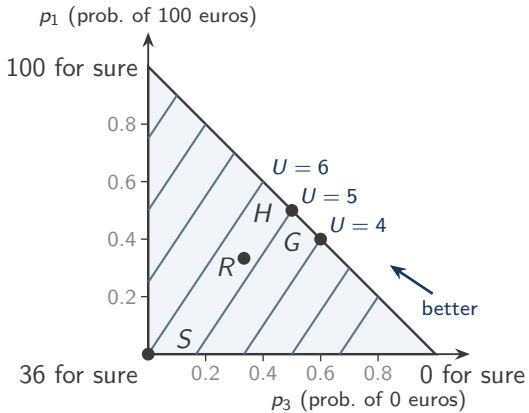
Expected utility theorem: if $\succsim$ on $\mathcal{L}$ is rational, continuous and satisfies independence, there are $u_1, \ldots, u_N$ such that for all L, L':

$$L \succsim L' \iff \sum_n u_n p_n \geq \sum_n u_n p'_n.$$

In the triangle: indifference curves are straight, parallel lines.

Conversely, every expected utility U satisfies independence (MWG Ex. 6.B.2).

MWG Prop. 6.B.3, (6.B.4), p. 176; proof pp. 176–178; Fig. 6.B.5, p. 175.



In pairs: reduce, then rank without computing

Outcomes 100, 36, 0 euros; her utilities 10, 6, 0.
$S = (0, 1, 0)$, $G = (0.4, 0, 0.6)$, $H = (0.5, 0, 0.5)$.

$K = (G, H;\ \frac{1}{2}, \frac{1}{2})$, $\quad K' = (S, H;\ \frac{1}{2}, \frac{1}{2})$

Per pair: three answers, three minutes.

(1) Reduce K.

(2) $U(K)$.

(3) K or K'? Which axiom?

Answers revealed in class. Try it in pairs before class.

MWG Defs. 6.B.2, 6.B.4; Exercise 6.B.1, p. 172 (strict version).

(optional) Expected utility form $\Rightarrow$ independence

Show: if U has the expected utility form, then $\succsim$ satisfies the independence axiom.

Hint: by Prop. 6.B.1, $U(\alpha L + (1-\alpha)L'') = \alpha\, U(L) + (1-\alpha)\, U(L'')$.

At home if not done in class; the answer is in the slides released after class.

MWG Exercise 6.B.2, p. 175.

Break

Ten minutes

Revealed in class

Before class: in each pair, what do the two lotteries have in common?

Expected utility is unique up to $\beta U + \gamma$, $\beta > 0$; u^2 flips her choice

utilities of (100, 36, 0 euros)	$U(G)$ $G = 40\,\%$ of 100	$U(S)$ $S = 36$ for sure	she picks
$u = (10, 6, 0)$	4	6	S
$2u + 3 = (23, 15, 3)$	11	15	S
$u^2 = (100, 36, 0)$	40	36	G

$\tilde{U}$ is another vNM utility for $\succsim \iff \tilde{U} = \beta U + \gamma$ with $\beta > 0$.

u^2 keeps the order of prizes, not lotteries:
unlike class 3, expected utility is cardinal.

MWG Prop. 6.B.2, pp. 173–174.

Money lotteries and risk aversion

MWG 6.C

A money lottery is a distribution function F; $U(F) = \int u(x)\, dF(x)$

$F(x)$ = probability that the payoff is at most x.

$U(F) = \int u(x)\, dF(x)$; with prizes x_n and probabilities p_n: $\sum_n p_n\, u(x_n)$.

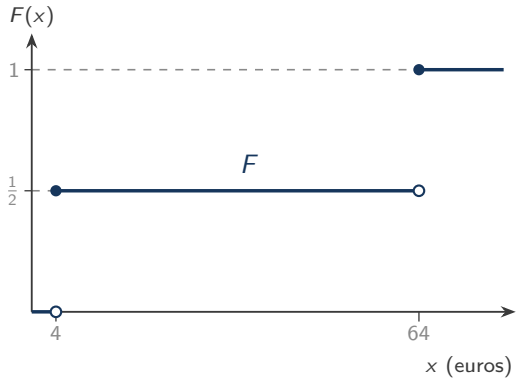
U: vNM utility, on lotteries.

u: Bernoulli utility, on money — increasing, continuous.

Her 10, 6, 0 for 100, 36, 0 euros: $u(x) = \sqrt{x}$.

MWG p. 183; (6.C.1) and fn. 12, p. 184; p. 185.

F: 4 or 64 euros, $\frac{1}{2}$ each



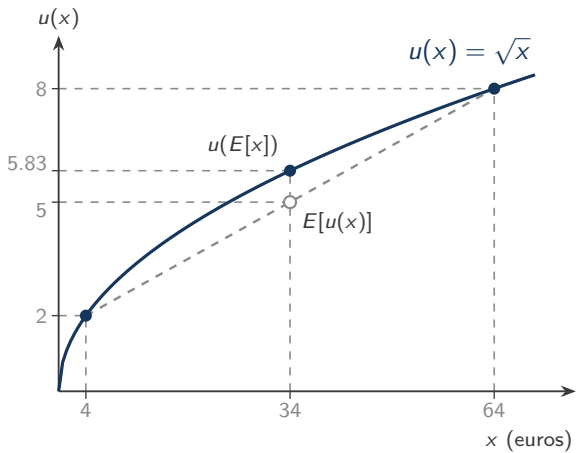
Risk averse $\Leftrightarrow$ u concave: the chord lies below the graph

Risk averse: for every F, the sure amount $\int x\, dF(x)$ is at least as good as F.

With expected utility: $\int u(x)\, dF(x) \leq u(\int x\, dF(x))$ for every F — Jensen: u concave.

Concave in money: a property of the function u, not convexity of preferences over bundles (classes 2–3). U is linear in probabilities.

MWG Def. 6.C.1 and (6.C.2), p. 185; Fig. 6.C.2, p. 186.



Certainty equivalent 25 euros, risk premium 9 euros: three steps

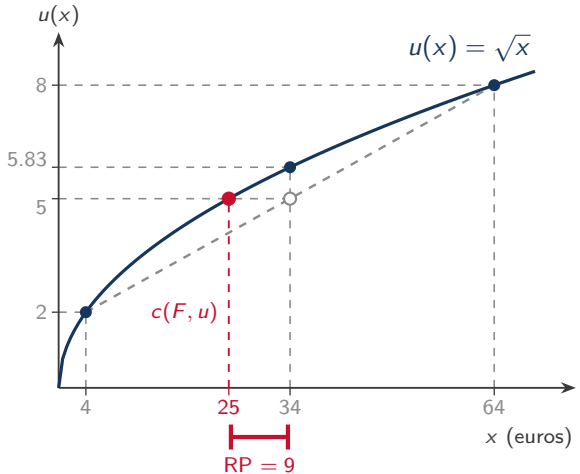
Certainty equivalent $c(F,u)$:

$u\Big(c(F,u)\Big) = \int u(x)\,dF(x).$

Risk premium: $\mathrm{RP} = E[x] - c(F,u)$.

1. $E[x] = \frac{1}{2}\cdot 4 + \frac{1}{2}\cdot 64 = 34$ euros
2. $E[u(x)] = \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 8 = 5$ utils
3. $c(F,u) = u^{-1}(5) = 25$ euros;
   $\mathrm{RP} = 34 - 25 = 9$ euros

MWG Def. 6.C.2(i), (6.C.3), p. 186: "some expected return is traded for certainty". RP as in Muñoz-Garcia, p. 357.



In pairs: one game, three people

Toss two coins. Both heads: 64 euros. Otherwise: nothing.

In pairs, one table per pair: four numbers per row. Four minutes.

	Bernoulli u	$E[x]$ euros	$E[u(x)]$ utils	$c(F,u)$ euros	RP euros
Ana	$\sqrt{x}$				
Ben	x				
Carla	x^2				

Answers revealed in class. Try it in pairs before class.

The three steps of slide 21.

One game, three people: a picture

Revealed in class

Do slide 22 first: the picture shows its answers.

(optional) Ana is richer: what is the game worth to her now?

Ana, $u(x) = \sqrt{x}$, has 36 euros plus slide 22's game (64 euros, probability $\frac{1}{4}$).

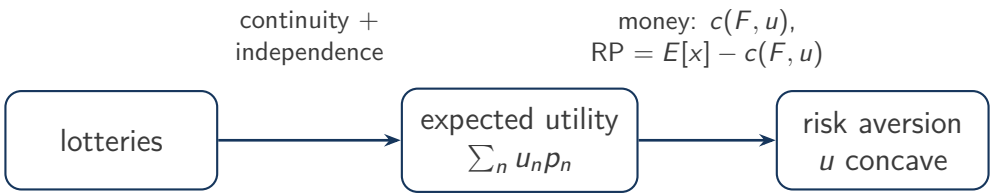
(1) $E[x]$, $E[u(x)]$ of her wealth.

(2) $c(F, u)$, and the game's value.

(3) RP.

At home if not done in class; the answers are in the slides released after class.

Today in one line



Tuesday 27 October: Uncertainty II: Arrow–Pratt measures, comparing risk aversion, insurance. First- and second-order stochastic dominance, mean-preserving spreads. Review.

Read MWG 6.C–6.D (or Muñoz-Garcia ch. 5) for Tuesday 27 October.